

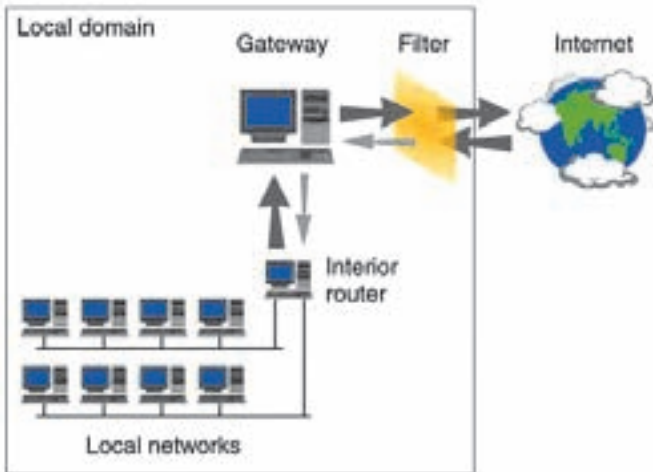
1.3 Firewalls

The internet is a gateway to sending and receiving limitless information. Like most gates, it has its share of intruders. And hence we have the gatekeepers – firewalls.

A firewall is part of a computer system or network designed to prevent unauthorised access while allowing verified and safe communications. It usually consists of a device or set of devices. It can also be implemented via hardware or software, or often a combination of both. Firewalls are most commonly used to prevent unauthorised internet users from accessing private networks, especially intranets. Each firewall has a different kind of authorisation criteria. Information entering or leaving the network through firewall that doesn't meet this standard is blocked.

Firewall techniques consist of four main types:

1. Packet filter: Packets are the most basic unit of data transfer between computers and networks. A packet



filter uses a set of user defined rules for identifying the source, destination address, protocol and port number. If a particular packet matches the rules, the filter either discards it or rejects it (also drops the packet, but sends